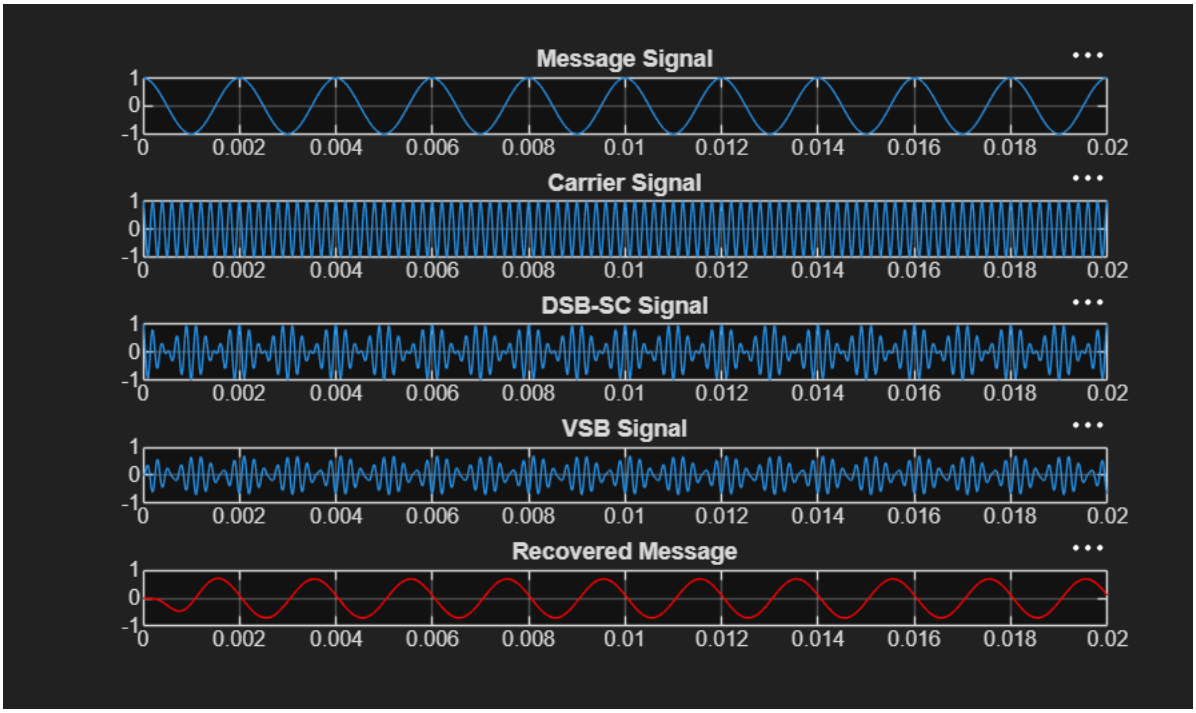
MATLAB-BASED MODELLING OF VSB MODULATOR AND

DEMODULATOR FOR ANALOG TELEVISION

COMMUNICATION APPLICATIONS

```
clc; clear;
close all;
% Parameters fs = 100000; % Sampling
Frequency t = 0:1/fs:0.02; % Time
Vector fm = 500; % Message Frequency
fc = 5000; % Carrier Frequency
% Message Signal m =
cos(2*pi*fm*t); %
Carrier Signal c =
cos(2*pi*fc*t);
% DSB-SC Modulation dsb =
m .* c;
% VSB Filter (Low-pass + High-pass Approximation)
[b,a] = butter(4, fc/(fs/2)); vsb
= filter(b,a,dsb);
% Coherent Demodulation demod = vsb
.* (2*cos(2*pi*fc*t));
% Recover Message
[b1,a1] = butter(4, 1000/(fs/2)); rec
= filter(b1,a1,demod); % Display
Results figure; subplot(5,1,1)
plot(t,m) title('Message Signal')
```

```
grid on subplot(5,1,2) plot(t,c)
title('Carrier Signal') grid on
subplot(5,1,3) plot(t,dsb)
title('DSB-SC Signal') grid on
subplot(5,1,4) plot(t,vsb) title('VSB
Signal') grid on subplot(5,1,5)
plot(t,rec,'r') title('Recovered
Message')
grid on
```